

Credit Card Default Prediction Using Machine Learning

1. Project Overview

Objective

The objective of this project is to develop a machine learning model capable of predicting whether a credit card customer will default on the next month's payment.

Early identification of potential defaulters helps financial institutions reduce credit risk, improve lending decisions, and optimize customer management strategies.

2. Dataset Information

Dataset

Default of Credit Card Clients Dataset

Records

30,000 customers

Target Variable

default.payment.next.month

0 = Non Default

1 = Default

Initial Features

LIMIT_BAL

SEX

EDUCATION

MARRIAGE

AGE

PAY_0 to PAY_6

BILL_AMT1 to BILL_AMT6

PAY_AMT1 to PAY_AMT6

3. Data Cleaning

Missing Values

No significant missing values were present.

Special PAY Status Values

The repayment status variables contained undocumented values:

-2 → No Consumption / No Bill Issued

0 → Paid On Time

These values were retained because they represent meaningful customer repayment behavior.

Removed Features

ID

The ID column had no predictive significance.

4. Exploratory Data Analysis

Key Findings

Education vs Default Rate

Education	Default Rate
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Graduate	19.23%
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High School	25.15%
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University	23.73%
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Others	5.69%
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Observation:

Customers with lower educational backgrounds exhibited higher default tendencies.

Marriage vs Default Rate

Marriage Status	Default Rate
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Married	23.47%
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Single	23.60%
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Others	20.92%
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Observation:

Marital status showed minimal impact on default behavior.

Gender vs Default Rate

Gender Default Rate

Female 20.77%

Male 24.16%

Observation:

Male customers demonstrated a slightly higher probability of default.

5. Repayment Behaviour Analysis

PAY variables showed the strongest relationship with default risk.

Key observations:

- Customers with no delay or on-time repayment had default rates below 20%.
- Default rate increased significantly as repayment delay increased.
- Delays of 2 months or more were strongly associated with future default.

This indicated that historical repayment behavior is the most important predictor of credit default.

6. Feature Engineering

To improve predictive performance, several new features were created.

Average Delay

Avg_Delay

Average repayment delay across repayment history.

Total Payment

Total_Payment

Sum of all payment amounts.

Credit Utilization

Credit_Utilised

=

Average Bill Amount / Credit Limit

Measures how much of the available credit is being used.

Payment Ratio

Pay_Ratio

=

Total Payment / Total Bill

Measures repayment efficiency.

7. Correlation Analysis

Strongest Positive Correlations

Feature	Correlation
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PAY_0	0.32
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Avg_Delay	0.281
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PAY_2	0.26
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Strongest Negative Correlations

Feature	Correlation
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LIMIT_BAL	-0.153
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Total_Payment	-0.102
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Insights:

- Higher repayment delays increase default probability.
 - Larger credit limits reduce default risk.
 - Customers making larger payments are less likely to default.
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8. Class Imbalance

Target Distribution

Non Default : 77.88%

Default : 22.12%

To handle imbalance, class weights and threshold tuning were applied during model development.

9. Feature Scaling

The following numerical features were standardized using StandardScaler:

LIMIT_BAL
AGE
Avg_Delay
Total_Payment
Credit_Utilised
Pay_Ratio

10. Model Development

Four machine learning models were trained and evaluated.

Logistic Regression

Metric	Score
Accuracy	68.93%
Precision	37.50%
Recall	60.74%
F1 Score	46.37%
ROC-AUC	71.04%

Decision Tree

(GridSearchCV Optimized)

Metric	Score
Accuracy	77.03%
Precision	48.36%
Recall	56.66%
F1 Score	52.18%
ROC-AUC	75.61%

Random Forest

(GridSearchCV Optimized)

Metric	Score
Accuracy	64.63%
Precision	35.81%
Recall	75.65%
F1 Score	48.61%
ROC-AUC	77.43%

Threshold tuning was performed to prioritize recall and improve identification of potential defaulters.

XGBoost (Best Model)

Metric	Score
Accuracy	77.00%
Precision	48.41%
Recall	60.96%
F1 Score	53.97%
ROC-AUC	77.84%

11. Model Explainability (SHAP)

SHAP analysis was performed to understand feature contributions.

Top Influential Features:

1. PAY_0
2. Avg_Delay
3. PAY_2
4. Total_Payment
5. LIMIT_BAL

SHAP summary plots demonstrated that repayment history was the primary driver behind model predictions.

12. Streamlit Deployment

A Streamlit application was developed to allow users to:

- Enter customer information
- Predict default probability
- Receive risk classification

Output:

High Risk of Default

or

Low Risk of Default

13. Conclusion

The project successfully developed an interpretable machine learning solution for predicting credit card defaults.

Among all evaluated models, XGBoost achieved the strongest balance between predictive performance and business relevance.

Key business findings include:

- Repayment history is the strongest predictor of default.
- Customers with higher repayment delays exhibit significantly higher default risk.
- Higher credit utilization increases risk.
- Larger payment amounts and credit limits reduce risk.

The developed system can support financial institutions in identifying high-risk customers and improving credit risk management.